

Joshua Peter

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🌐 Website

Summary

Fourth-year mechanical engineering undergraduate student with experience in people/project management, CAD/BIM/FEA modeling, and basic Python/MATLAB programming seeking an internship in sustainability-conscious power systems.

Education

B.S. Mechanical Engineering

Expected May 2027

University of Houston, Cullen College of Engineering – Houston, TX

GPA: 3.9 (Dean's List, Fall 2023–Present)

Relevant Coursework: Finite Element Methods, Design of Machine Elements, Fluid Mechanics, Materials Science

Work Experience

Schweitzer Engineering Laboratories, Inc. (SEL)

Houston, TX

Engineering Intern, Substation Engineering Services

Jan. 2025–Present

- Design various types of foundations and steel structures for both brown- and greenfield projects, performing site-specific calculations and modeling using software like RISA Foundation, RISA 3D, and LPILE
- Consult civil engineering standards (ASCE 7, IBC) to furnish calculation briefs for foundation and steel structure designs
- Develop specialized internal Excel templates using AI-assisted “vibe coding” in Visual Basic for Applications (VBA), creating instructional videos and typeset job aids to aid in coaching team members on their use
- Perform non-technical QA/QC reviews on drawing packages to be submitted to customers

Bean Here Coffee LLC

Missouri City, TX

Part-Time Senior/Managing Barista

May 2021–Aug. 2024

- Administered over 20 employees on-staff and onboarded more than 5 new hires with a self-implemented training regimen
- Proposed new workflow recommendations through the institution of a barista “best practices” handbook and innovative coffee quality assurance resources

Relevant Projects

Design-Build-Test (DBT) Project

University of Houston

Project Manager

Spring 2025

- Served as project manager for a semester-long iteration of the engineering design process, directing a team of 4 other students to build a mechanical sorting device subject to weight, size, and time constraints
- Interfaced between instructors and project team, acting as a point of contact and facilitating routine design charrettes with all stakeholders
- Developed a detailed work breakdown structure to tackle deliverables, leveraging organizational tools like Gantt charts and weighted decision matrices, and software like ClickUp

Leadership & Professional Affiliations

Vice President Operations

May 2026–Present

American Society of Mechanical Engineers @ The University of Houston (ASME-UH)

Houston, TX

Student Mentor

Sep. 2025–Present

American Society of Mechanical Engineers @ The University of Houston (ASME-UH)

Houston, TX

Mentor 3 underclassmen in engineering professionalism, offering resume critiques/revisions, encouraging participation in ASME-UH events, and providing tutoring for previously completed courses

Skills & Certifications

Autodesk Inventor/Revit/AutoCAD · RISA Foundation/3D · LPILE · COMSOL · Python · MATLAB · Mathcad · ~~Blender~~ · Spanish (Intermediate) · Malayalam (Intermediate) · TIG Welding · American Red Cross Adult First Aid/CPR/AED (2025–2027) · PCEPTM – Certified Entry-Level Python Programmer (2023) · OSHA 10-Hour Construction Safety and Health (2021)